

## ***An open-label study on the effect of flax seed powder (*Linum usitatissimum*) supplementation in the management of diabetes mellitus.***

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Diabetes mellitus is characterized by hyperglycemia and associated with aberrations in the metabolism of carbohydrate, protein, and lipid that result in development of secondary complications. Extensive studies have indicated that nutritional therapy plays a pivotal role in the controlling or postponing of development of these secondary complications. Several functional foods have been shown to possess hypoglycemic and hypolipidemic properties. Flax seed (FS) is a functional food that is rich in omega 3 fatty acids and antioxidants and is low in carbohydrates. In exploratory studies, FS was incorporated in recipes, which resulted in a reduction in the glycemic index of the food items. These observations prompted us to investigate the efficacy of FS supplementation in type 2 diabetics (n = 29). Subjects were assigned to the experimental (n = 18) or the control group (n = 11) on the basis of their desire to participate in the study. The experimental group's diet was supplemented daily with 10 g of FS powder for a period of 1 month. The control group received no supplementation or placebo. During the study, diet and drug intake of the subjects remained unaltered. The efficacy of supplementation with FS was evaluated through a battery of clinico-biochemical parameters. Supplementation with FS reduced fasting blood glucose by 19.7% and glycated hemoglobin by 15.6%. A favorable reduction in total cholesterol (14.3%), triglycerides (17.5%), low-density lipoprotein cholesterol (21.8%), and apolipoprotein B and an increase in high-density lipoprotein cholesterol (11.9%) were also noticed. These observations suggest the therapeutic potential of FS in the management of diabetes mellitus.